



Biomechanical Analysis of Locomotion and Morphological Adaptations in Snakes from Distinct Habitats

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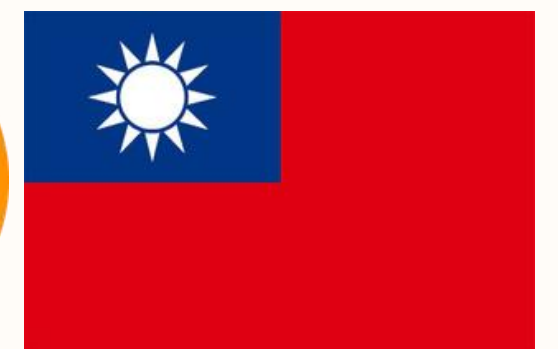


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Introduction

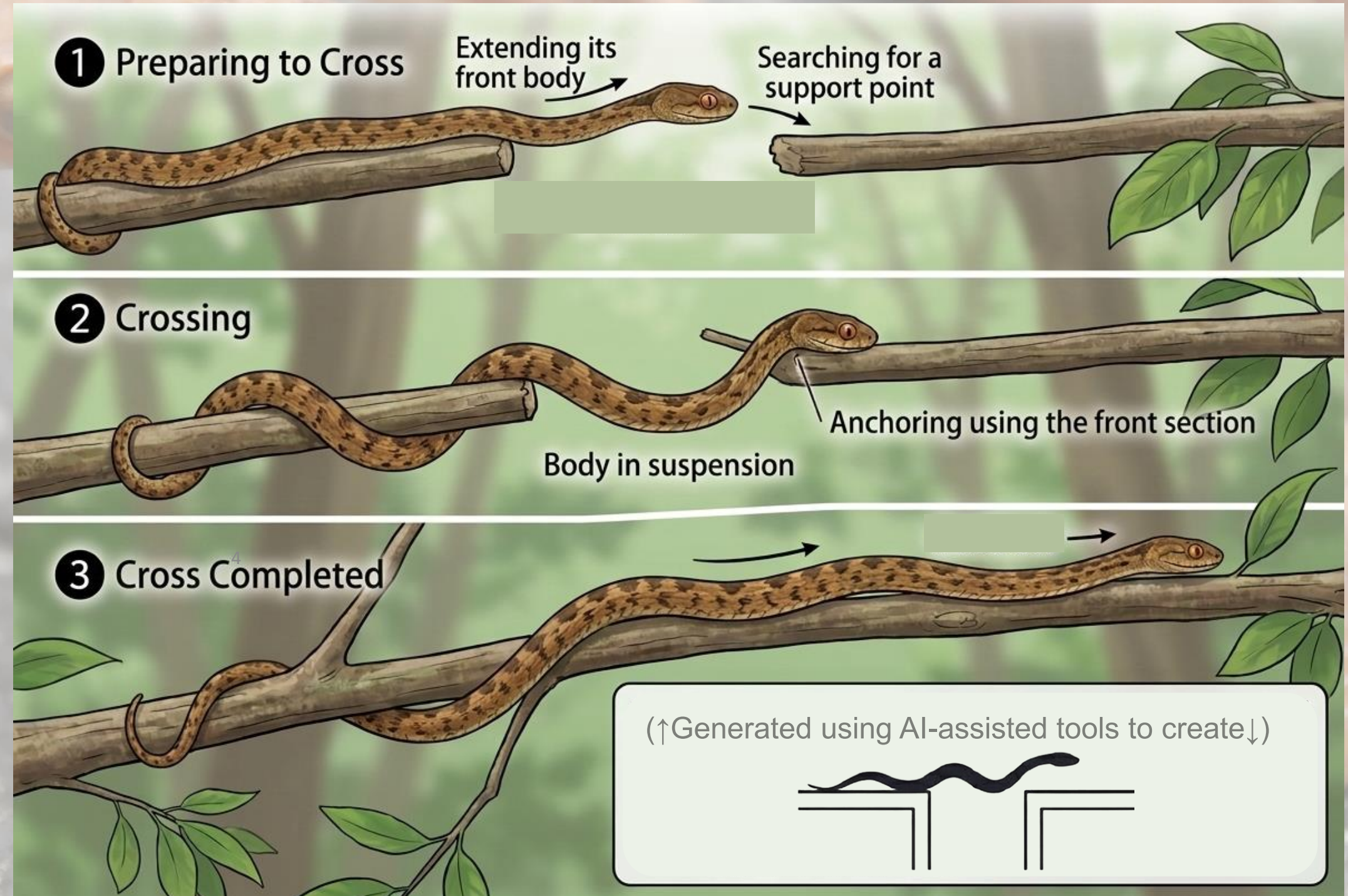
Motivation



Observation



Question



Objective

01

Body dimensions

02

**Vertical extention
ability**

03

**Horizontal reaching
ability**

Hypotheses

- **Arboreal snakes possess longer tail ratios to facilitate climbing.**
- **Arboreal species are superior to terrestrial species in vertical and horizontal locomotion.**

Research Species

Pantherophis guttatus

Habitat : North American

Habitat Type : Ground Dwelling

Food : Rodents, Birds

Body Length : 97cm



Heterodon nasicus

Habitat : North American

Habitat Type : Ground Dwelling

Food : Rodents, Lizards

Body Length : 47cm



Boa constrictor

Habitat : Central and South America

Habitat Type : Arboreal

Food : Mammals, Birds

Body Length : 150cm



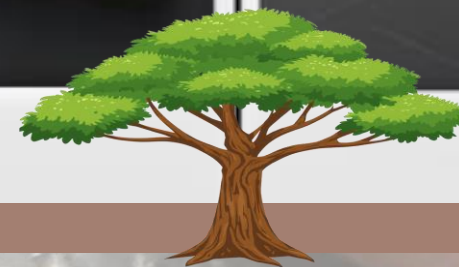
Boiga kraepelini

Habitat : Taiwan and China

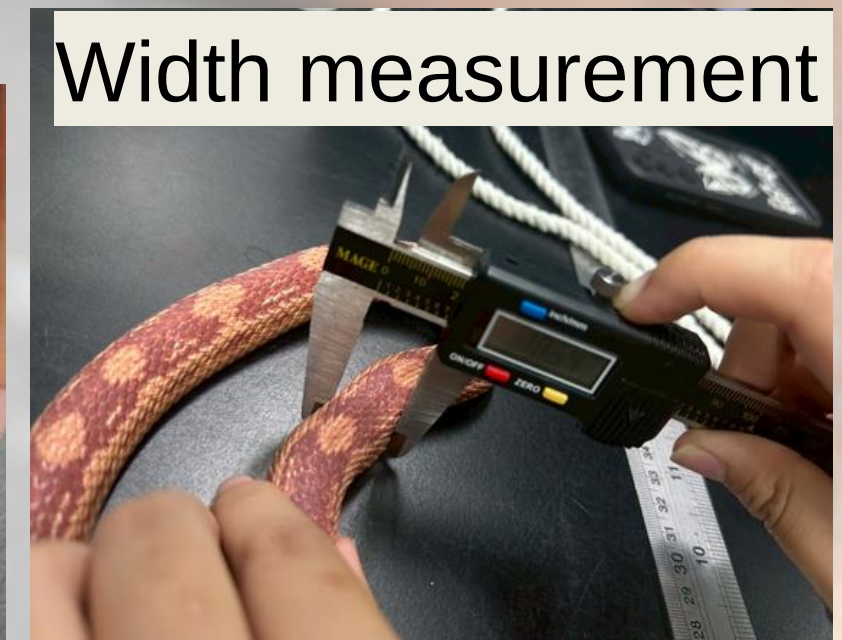
Habitat type : Arboreal

Food : Rodents, Birds

Body Length : 86cm



Method



Body Division:

-Head -Trunk -Tail (from vent to tail tip)

Data Processing: Measurements were normalized to body part percentages for standardized species comparison.

Method

Vertical extension



Horizontal reaching

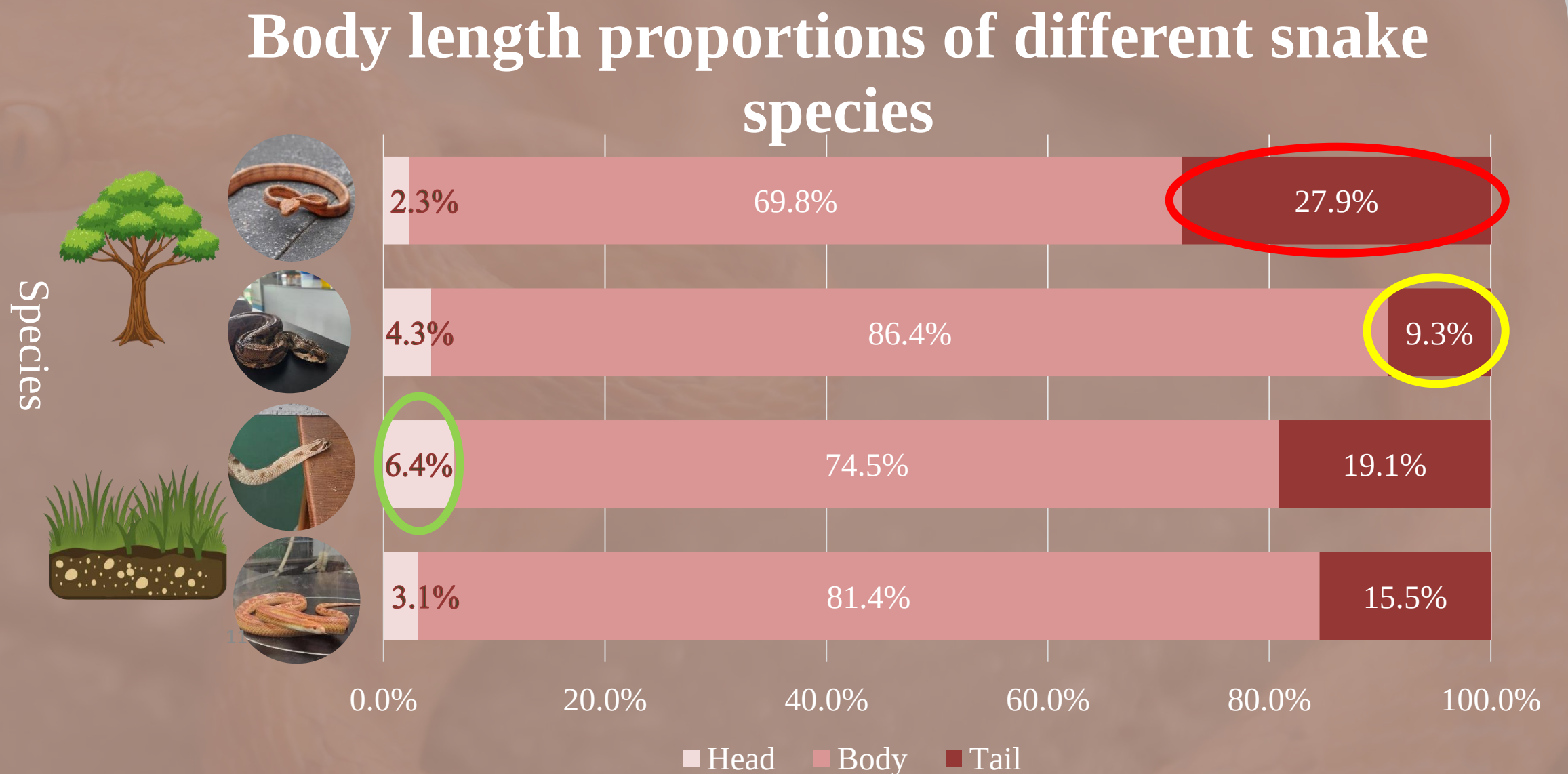




Result & Discussion

1. Body Dimensions

Species	Picture	Total length (cm)
<i>Boiga kraepelini</i>		86
<i>Boa constrictor</i>		150
<i>Heterodon nasicus</i>		47
<i>Pantherop his guttatus</i>		97

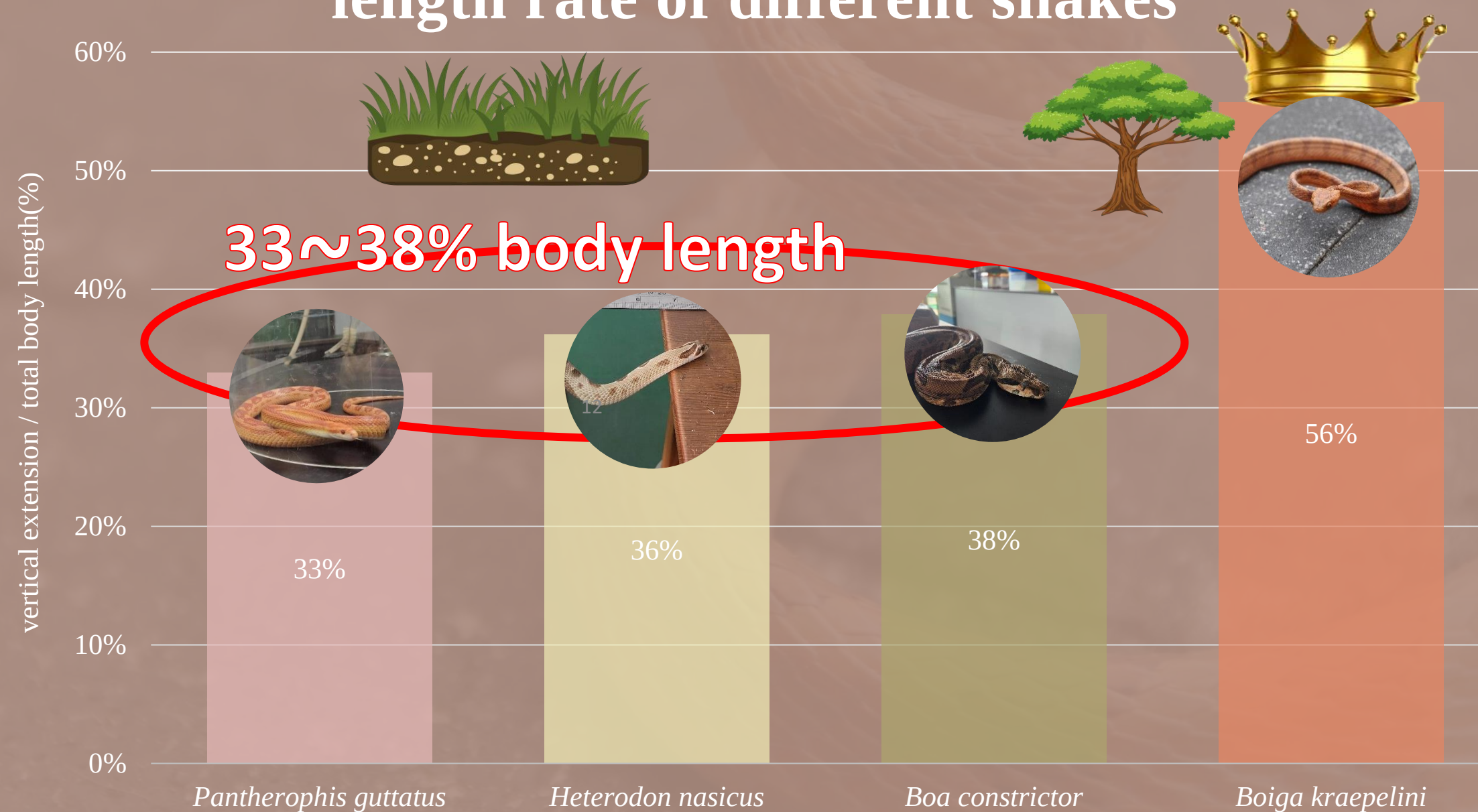


- *Boiga kraepelini*: Highest tail proportion (27.9%).
- *Heterodon nasicus*: Largest head proportion (6.4%).
- *Boa constrictor*: Shortest tail proportion (9.3%), but greatest total body length.

2. Vertical Extension

Best: *Boiga kraepelini* (56%) Other: between 33% and 38%

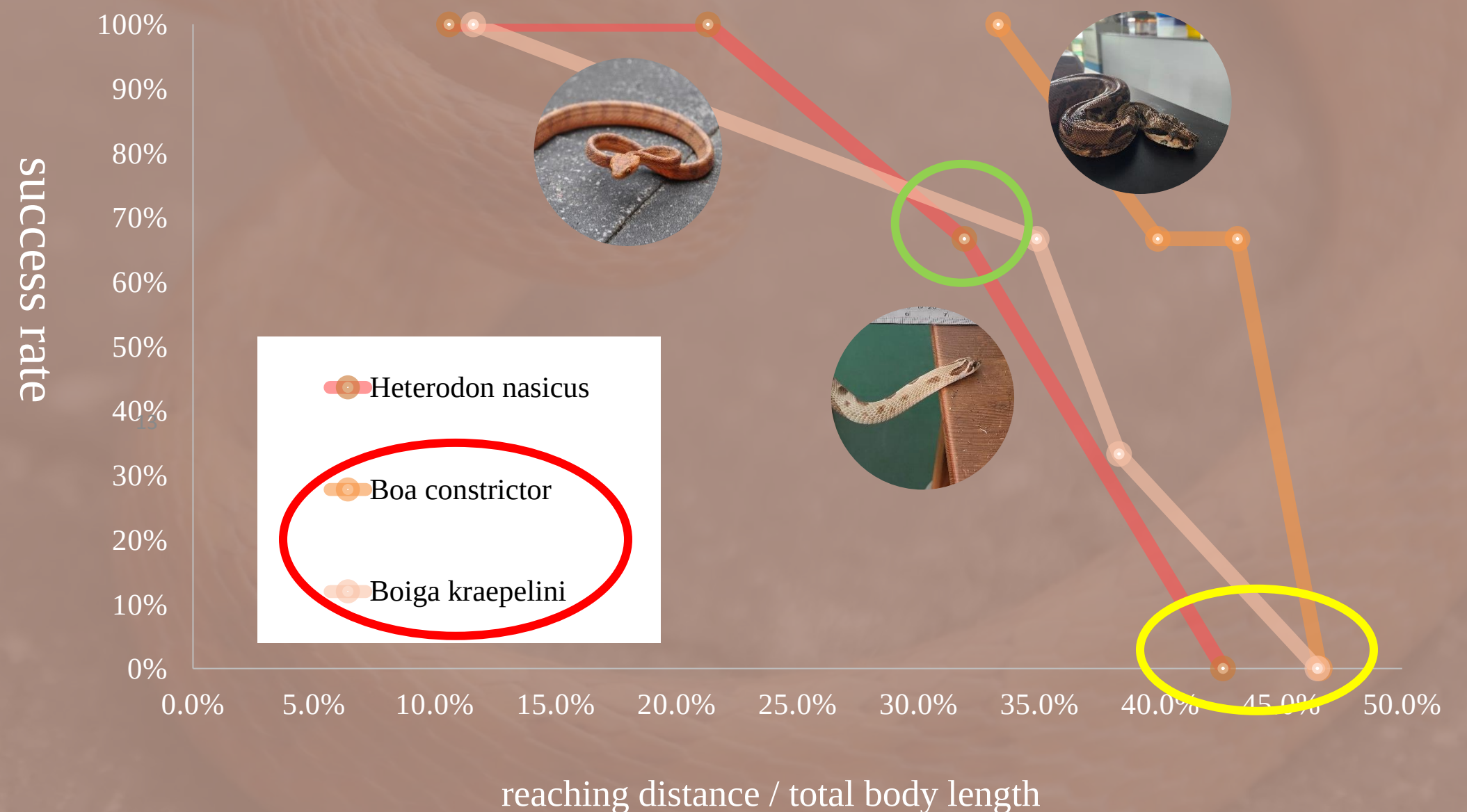
Largest vertical extension / total body
length rate of different snakes



3. Horizontal Reaching

- Success rates of arboreal decline gradually.
- *H. nasicus* drops sharply beyond a 32% extension.
- All species' results limit of ~43% body length.

Horizontal reaching success rate of different snakes

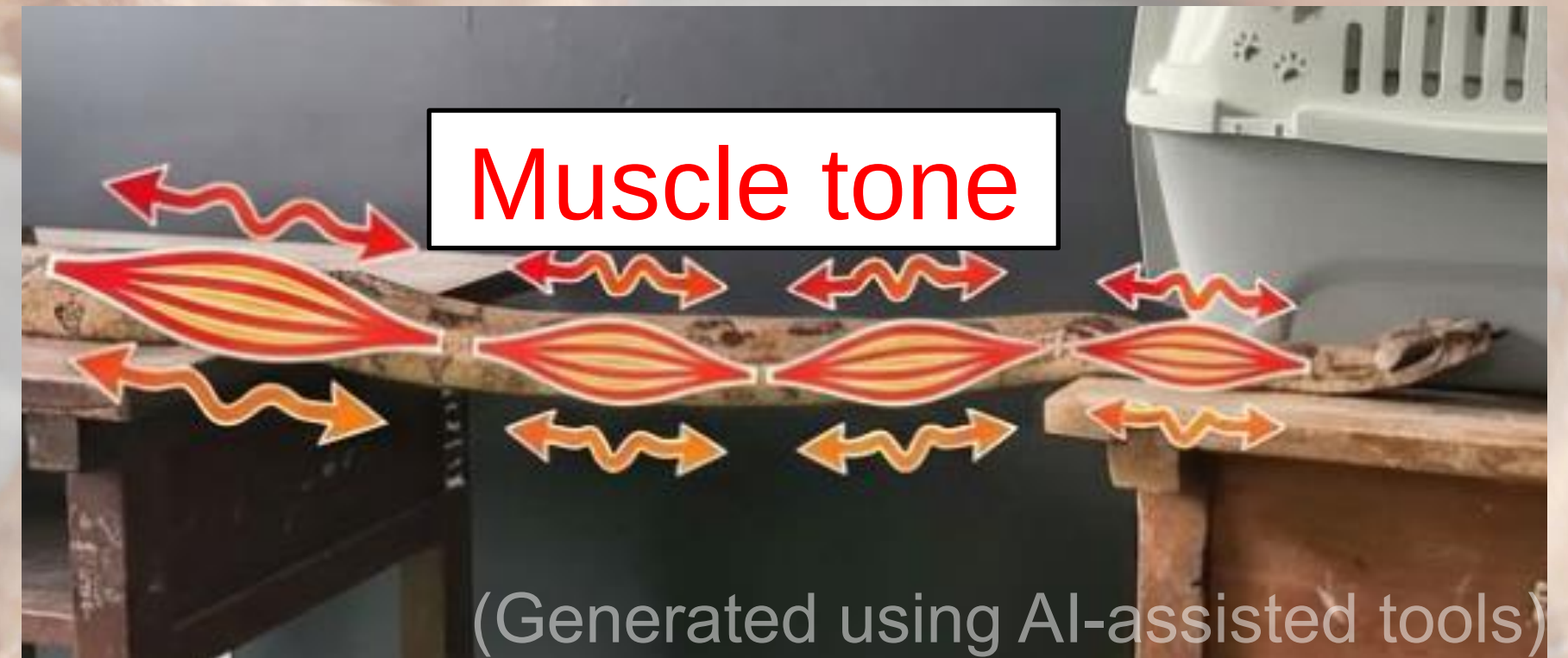
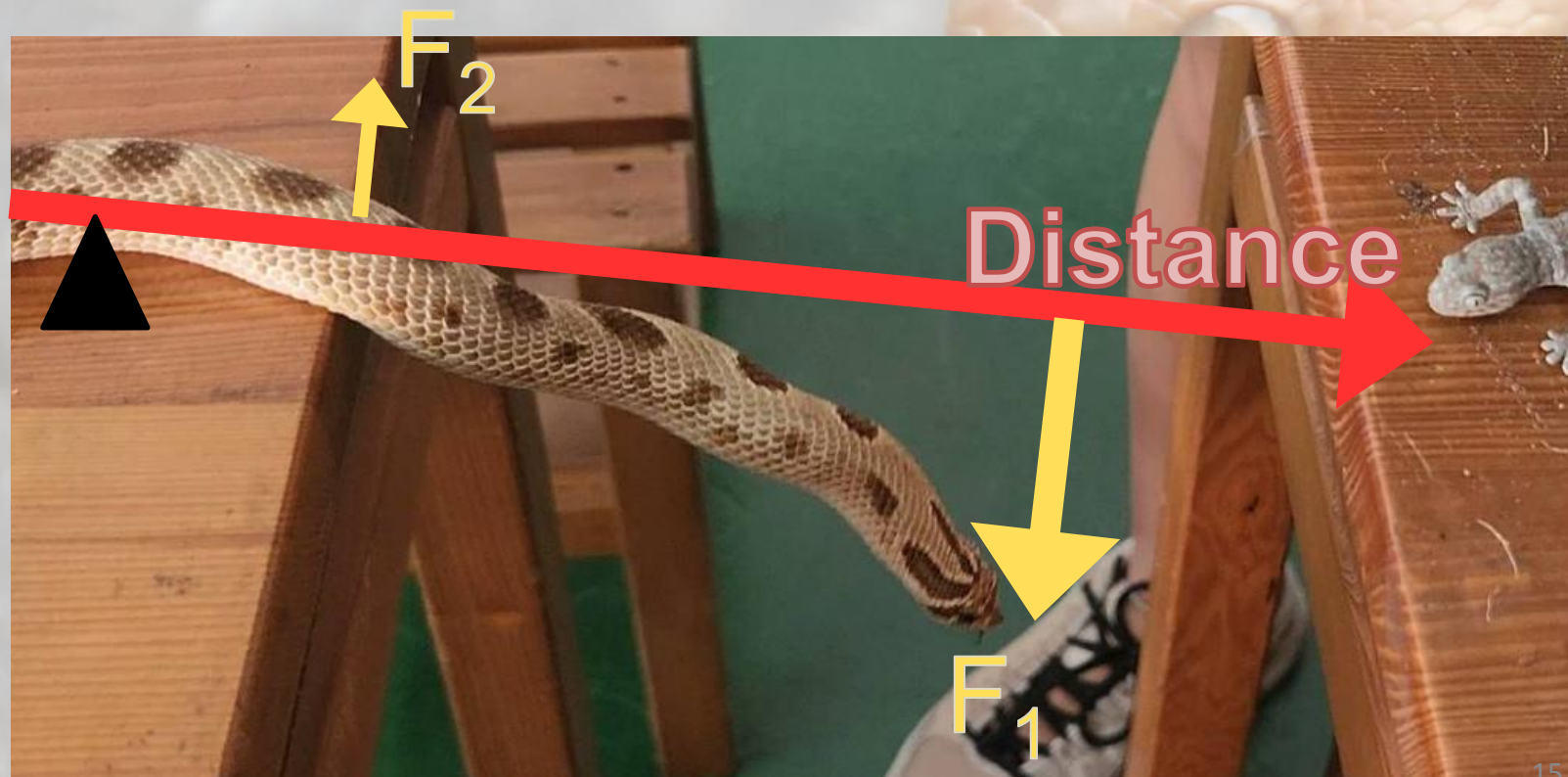




Conclusion

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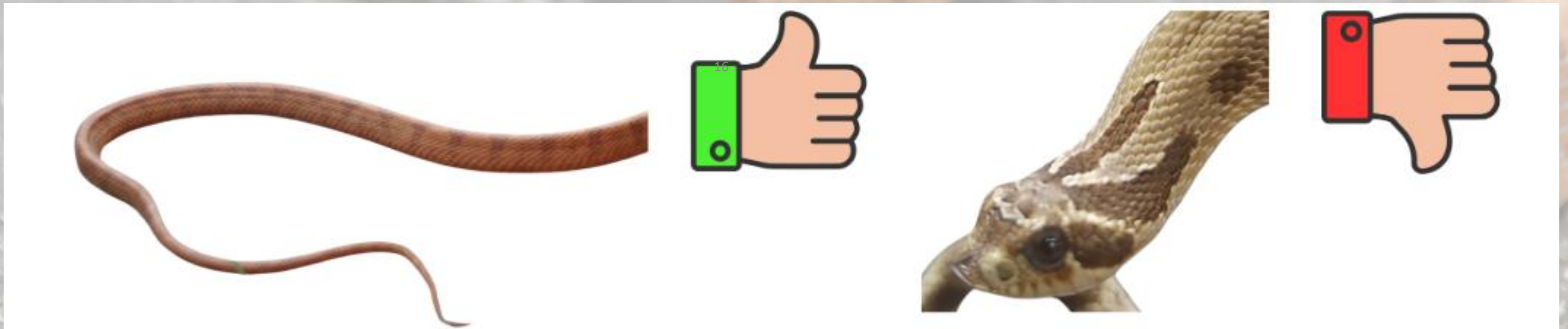
1. Mechanical Analysis of Horizontal Reaching



- $\tau = F \times d$ (F_1 = head weight , F_2 = counterweight)
→ Muscular strength can compensate for morphological disadvantages (e.g., a short tail).

2. Interpretation of Morphological Trade-offs

- **Long tail**→ a posterior anchor and counterweight.
- **Heavy head**→ greater anterior torque.



3. Differentiation of Locomotor Strategies

B. constrictor has superior muscular tension to make up for its short tail and massive body.





Future potential

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Scientifically Informed Coexistence

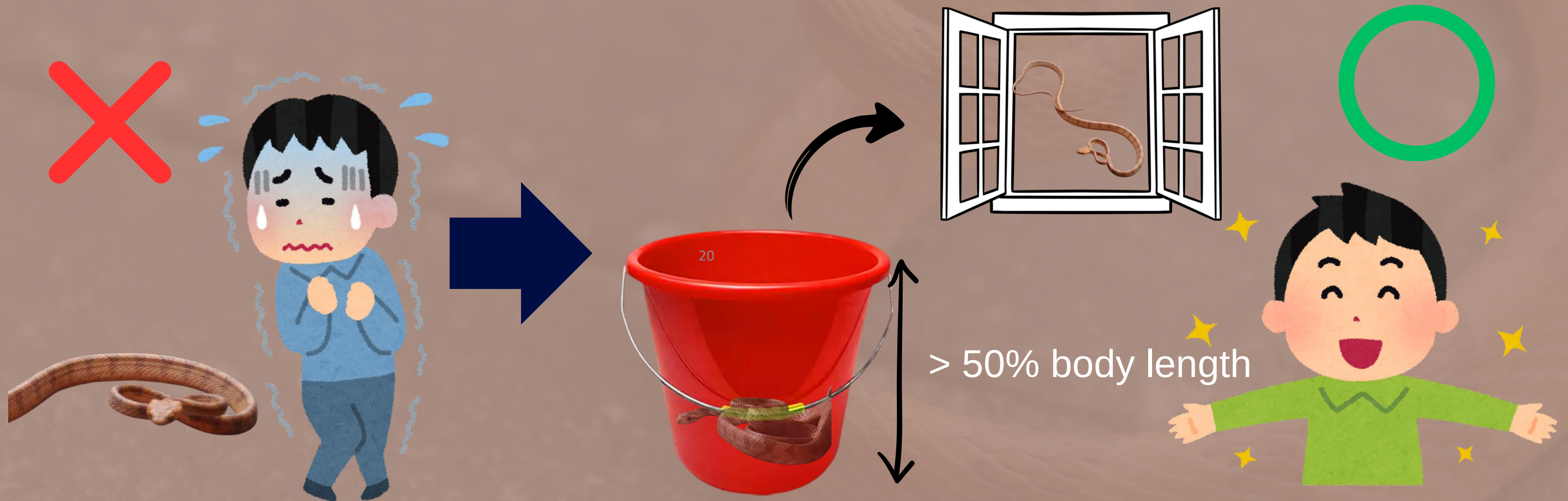


**Provide a scale for husbandry and
relocate snakes back in the wild**

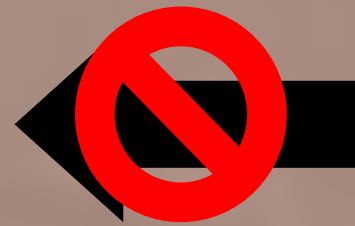
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Design habitat barriers

Provides a scale for husbandry and relocate snakes back in the wild



Design habitat barriers



> 43% body length

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Future Work



Verifying the 43% biomechanical constant across diverse species

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Researching scale micro-structures

References

1. Sheehy, Coleman M., et al. (2016). The evolution of tail length in snakes associated with different gravitational environments.
2. Greg Byrnes, Bruce C Jayne (2012). The effects of three-dimensional gap orientation on bridging performance and behavior of brown tree snakes (*Boiga irregularis*).
3. Bruce C Jayne, et al. (2015). Why arboreal snakes should not be cylindrical: body shape, incline and surface roughness have interactive effects on locomotion.

Thank You!

